Devnotes-ClpXP research notebook

Status	Shipped
Tags	
Researchers	Ⓨ Yen-Yu Hsu
Keywords	
Summary	The research focuses on integrating the ClpXP protease into the PURE system for effective temporal regulation of protein production. This integration aims to address limitations in controlling system dynamics, particularly the degradation of proteins to maintain homeostasis and regulate cell cycle behaviors. The notebook includes detailed experimental setups and results from various bulk reactions conducted over several months, highlighting the performance of different protein constructs and conditions.

Overview

*Summary of the research or experiment, particularly the **why** of the work.*

The Control Module of the Developer Cell Project focuses on achieving temporal regulation of protein production using the PURE system. While the PURE cytosol lacks proteases—offering an advantage in some contexts—it also presents a limitation for controlling system dynamics. For instance, when a sensor in the PURE cytosol is activated, it may remain permanently on since the output reporter protein is not degraded. As synthetic cell researchers advance toward more complex modules and cells, the ability to regulate protein degradation becomes essential for functions like level matching, homeostasis, and time-dependent behaviors, such as cell cycle regulation. To address this, we will integrate the ClpXP protease, enabling the continuous degradation of targeted proteins.

Contents

Background

Information and resources necessary for context; include links to other research notebooks as necessary.

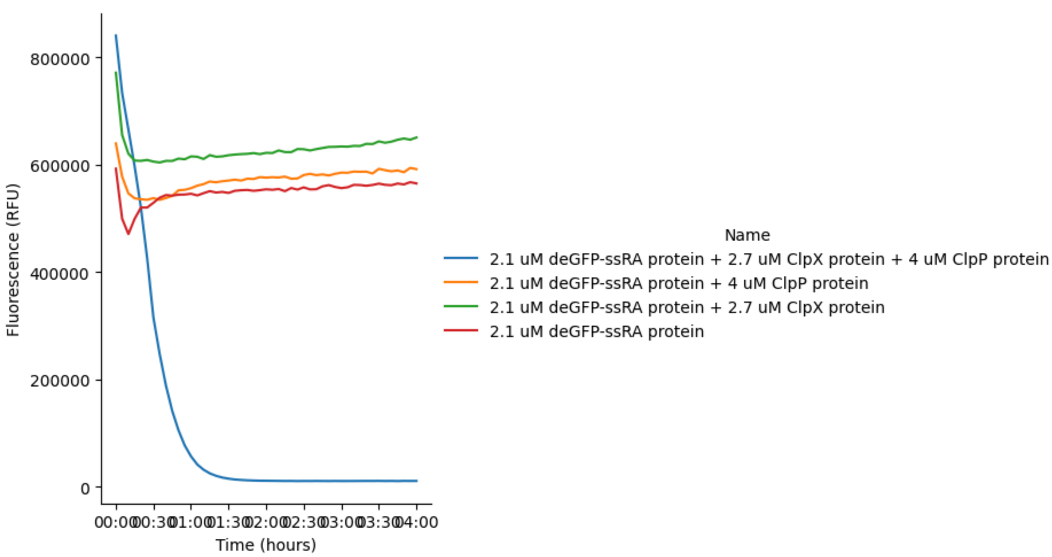
ClpXP is a well-characterized protease composed of two subunits: the ClpX ATPase and the ClpP peptidase. Its functionality has been successfully demonstrated within the PURE system. ClpXP specifically targets proteins that are tagged with a C-terminal ssrA tag, enabling selective degradation of only the desired proteins while avoiding the indiscriminate degradation of essential cytosolic machinery. Additionally, ClpXP's requirement for ATP further motivates its integration with the energy module, ensuring that both protein production and degradation can be efficiently managed to maintain optimal protein yield.

Notebook

@November 20, 2025

Bulk Reactions:

	Sample 1 (H1	Sample 2	Sample 3	Control
Purified deGFP-ssRA (1.15mg/ml) (41.2 µM)	0.5	0.5	0.5	0.5
Purified ClpP (1.94 mg/ml)(79.9µM)	0.5	0.5	0	0
Purified ClpX (2.55mg/ml)(53.7µM)	0.5	0	0.5	0
PURE	7.5	7.5	7.5	7.5
H2O	1	1.5	1.5	2
Total	10	10	10	10

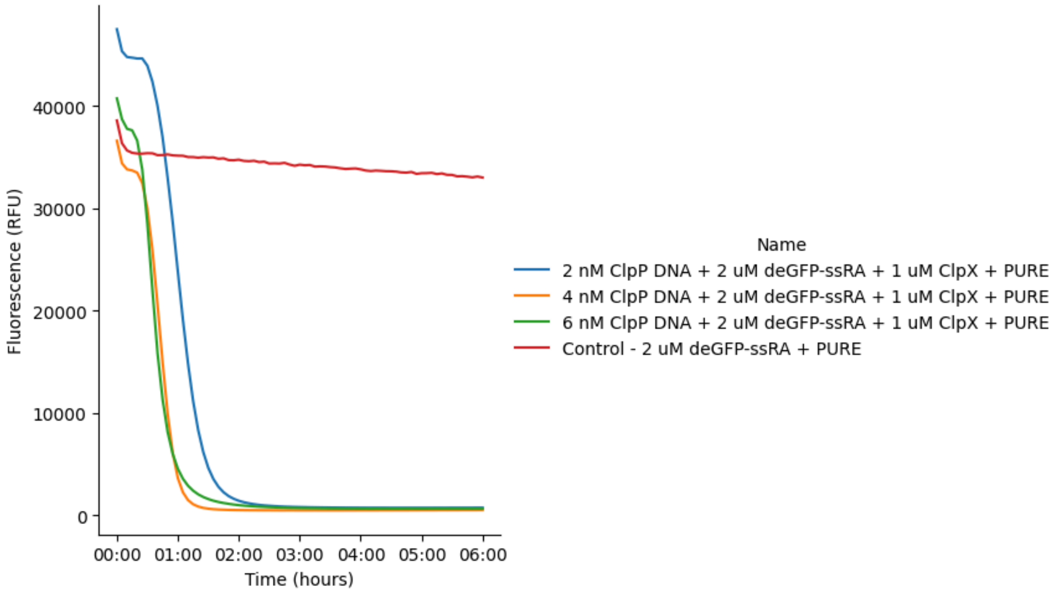


@September 19, 2025

Bulk Reactions:

	Sample 1	Sample 2	Sample 3	Control
pOpen-pT7-ClpP-cHis (40 nM)	0.5	1	1.5	0
Purified deGFP-ssRA (1.15mg/ml) (41.2 µM)	0.5	0.5	0.5	0.5
Purified ClpX (2.55mg/ml)(53.7µM)	0.2	0.2	0.2	0
PURE	7.5	7.5	7.5	7.5

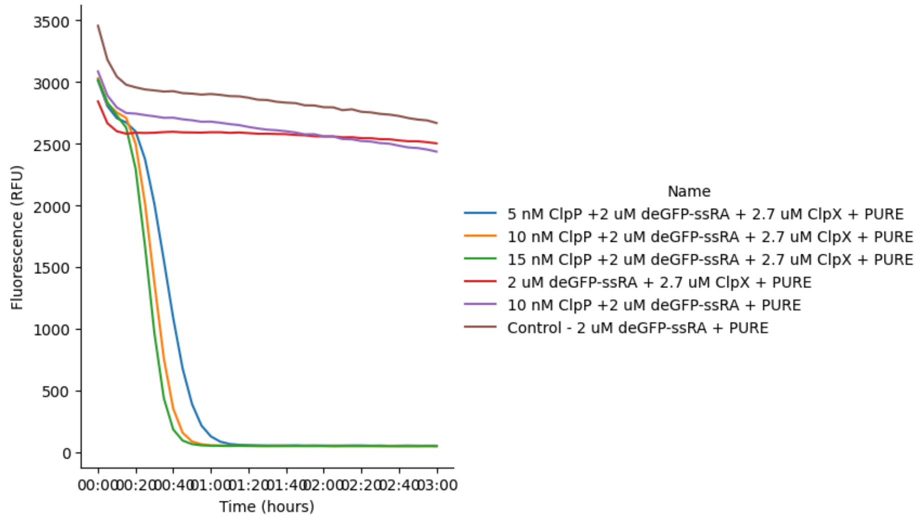
H2O	1.3	0.8	0.3	2
Total	10	10	10	10



@September 26, 2025

Bulk reactions:

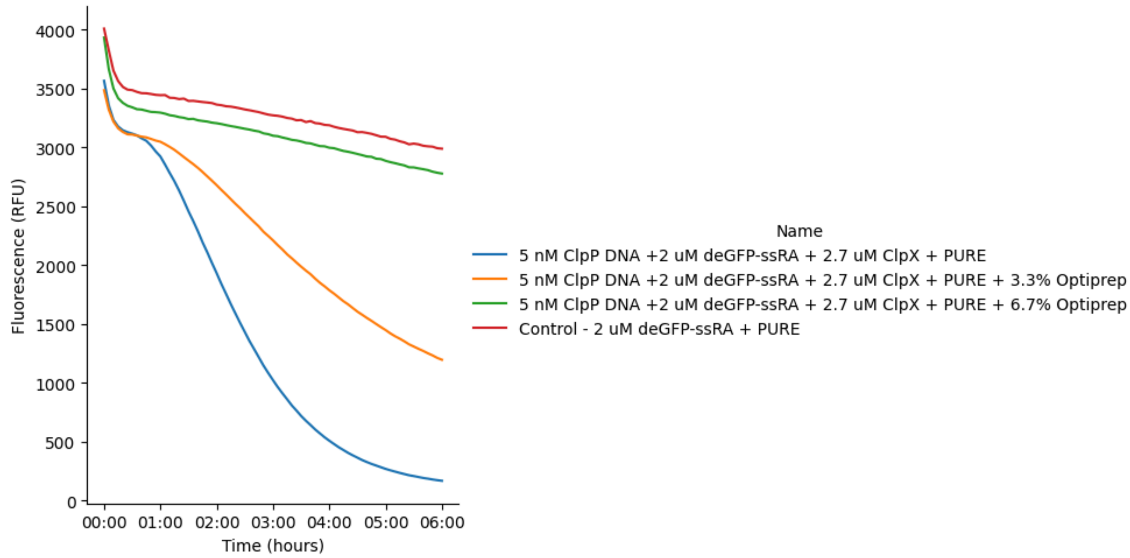
	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Control
pOpen-pT7-ClpP-cHis (47.85 ng/ul)	0.5	1	1.5	0	1	0
Purified deGFP-ssRA (1.15mg/ml) (41.2 µM)	0.5	0.5	0.5	0.5	0.5	0.5
Purified ClpX (2.55mg/ml) (53.7µM)	0.5	0.5	0.5	0.5	0	0
PURE	7.5	7.5	7.5	7.5	7.5	7.5
H2O	1	0.5	0	1.5	2	2
Total	10	10	10	10	10	10



@September 29, 2025

Bulk reactions:

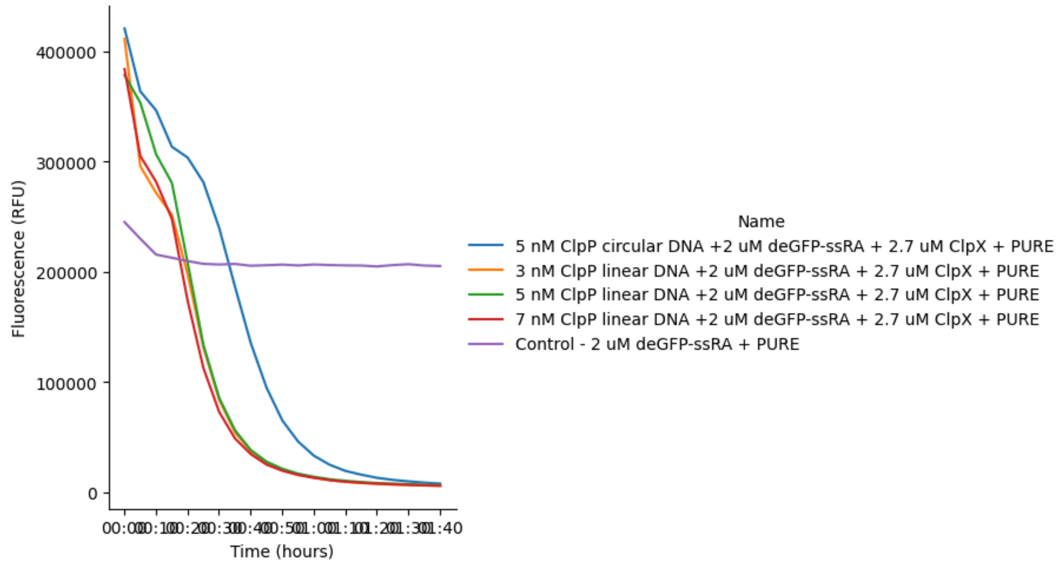
	Sample 1	Sample 2	Sample 3	Control
pOpen-pT7-ClpP-cHis (100 nM)	0.5	0.5	0.5	0
Purified deGFP-ssRA (1.15mg/ml)(41.2 µM)	0.5	0.5	0.5	0.5
Purified ClpX (2.55mg/ml)(53.7µM)	0.5	0.5	0.5	0
PURE	7.5	7.5	7.5	7.5
Optiprep	0	0.33	0.67	0
H2O	1	0.67	0.33	2
Total	10	10	10	10



@October 1, 2025

Bulk Reactions:

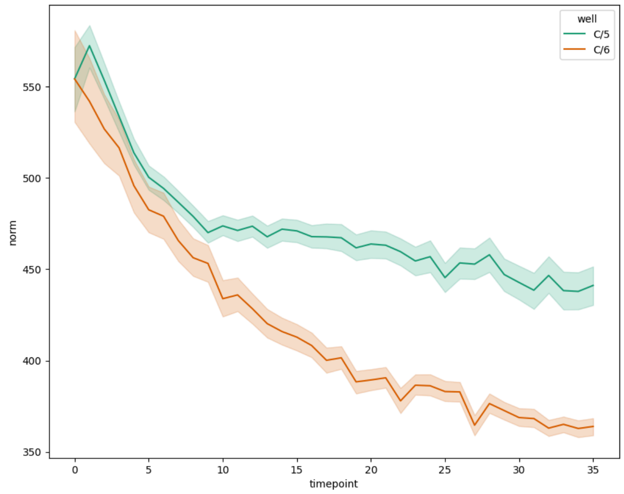
	Sample 1	Sample 3	Sample 2	Sample 4	Control
pOpen-pT7-ClpP-cHis (47.85 ng/ul)	0.5	x	x	x	x
Linear pOpen-pT7-ClpP-cHis (47.85 ng/ul)	x	0.3	0.5	0.7	x
Purified deGFP-ssRA (1.15mg/ml) (41.2 µM)	0.5	0.5	0.5	0.5	0.5
Purified ClpX (2.55mg/ml) (53.7µM)	0.2	0.2	0.2	0.2	0.2
PURE	7.5	7.5	7.5	7.5	7.5
H2O	1.3	1.5	1.3	1.1	1.8
Total	10	10	10	10	10

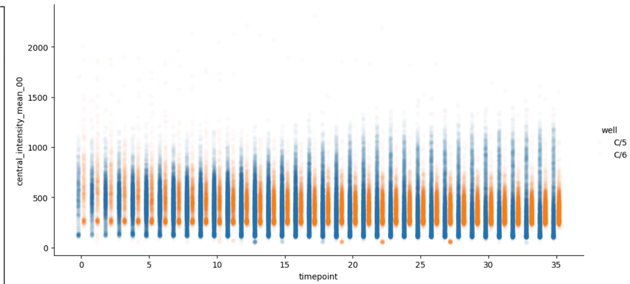


@October 3, 2025

Cells:

	Sample	Control
Linear pOpen-pT7-ClpP-cHis (47.85 ng/ul)	1.5	x
Purified deGFP-ssRA (1.15mg/ml)(41.2 µM)	0.3	0.3
Purified ClpX (2.55mg/ml)(53.7µM)	0.6	0.6
PURE	22.5	22.5
Optiprep	x	x
2M Sucrose	4.5	4.5
H2O	0.6	2.1
Total	30	30

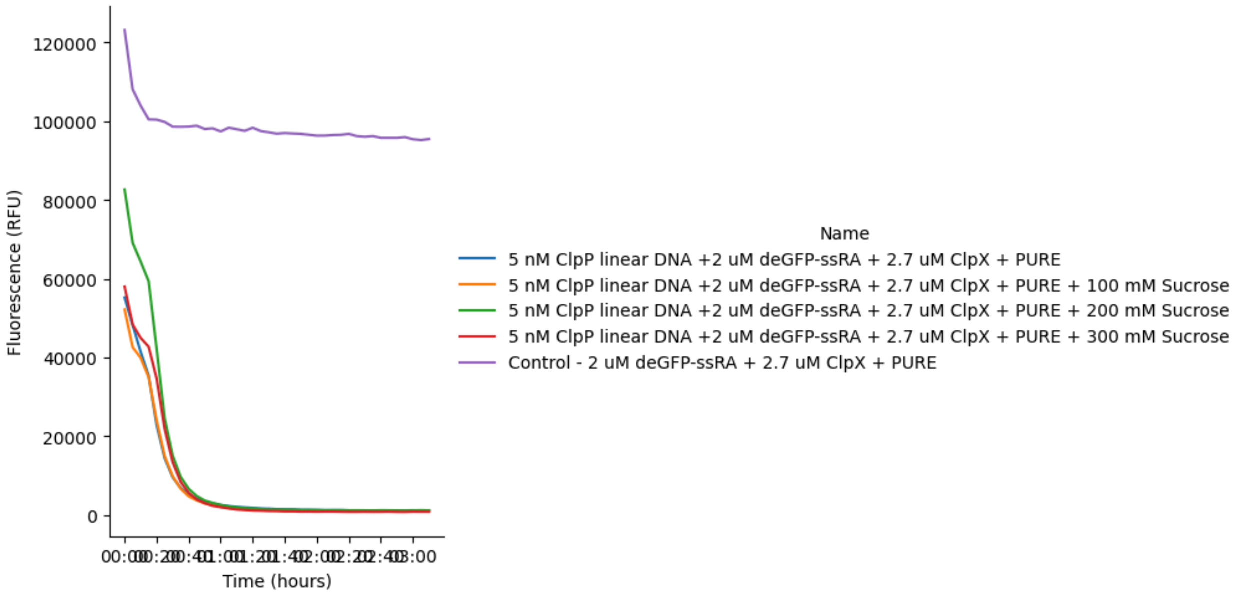




@October 8, 2025

Bulks:

	Sample 1	Sample 2	Sample 3	Sample 4	Control
pOpen-pT7-ClpP-cHis (47.85 ng/ul)	0.5	0.5	0.5	0.5	0
Purified deGFP-ssRA (1.15mg/ml) (41.2 µM)	0.1	0.1	0.1	0.1	0.1
Purified ClpX (2.55mg/ml) (53.7µM)	0.2	0.2	0.2	0.2	0.2
PURE	7.5	7.5	7.5	7.5	7.5
2M Sucrose	0	0.5	1	1.5	0
H2O	1.7	1.2	0.7	0.2	2.2
Total	10	10	10	10	10



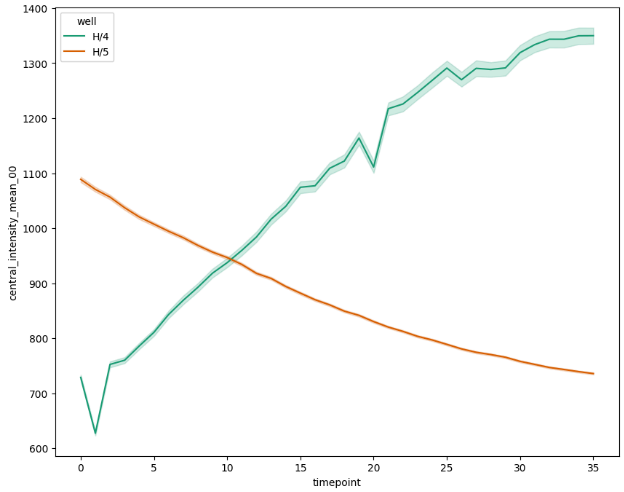
Osmolarity:

Purified GFP-ssrA: 2570 mOsm

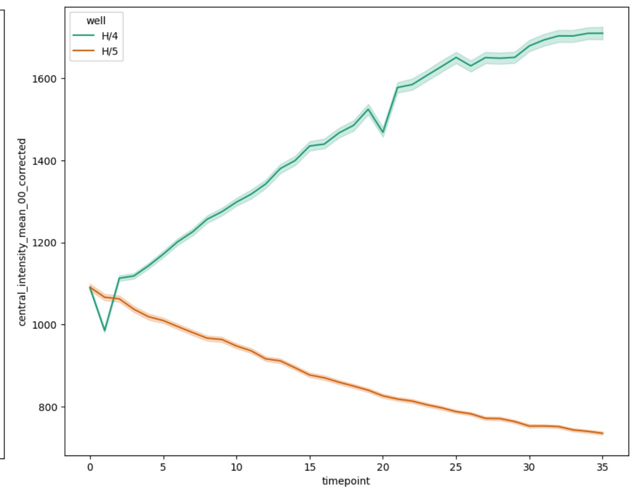
Purified ClpX: 2480 mOsm

Cells:

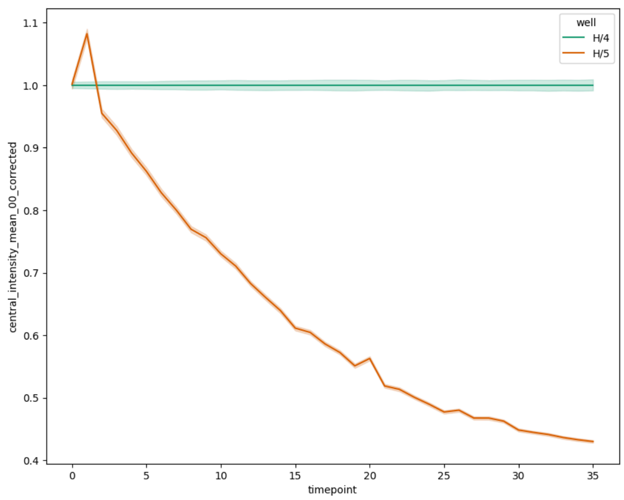
	Sample (H5)	Control (H4)
Linear pOpen-pT7-ClpP-cHis (47.85 ng/ul)	1.5	x
Purified deGFP-ssRA (1.15mg/ml)(41.2 µM)	0.3	0.3
Purified ClpX (2.55mg/ml)(53.7µM)	0.6	0.6
PURE	22.5	22.5
2M Sucrose	3	3.5
H2O	2.1	3.1
Total	30	30



Liposomes fluorescence overtime. The first image was captured 1.5 hr after the liposomes are made.



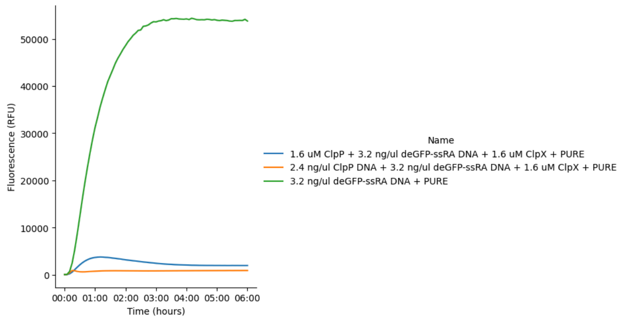
Background subtracted

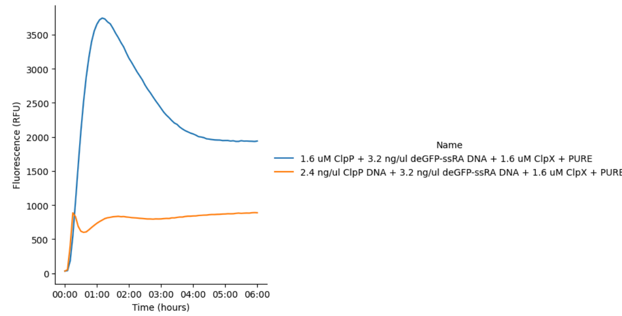


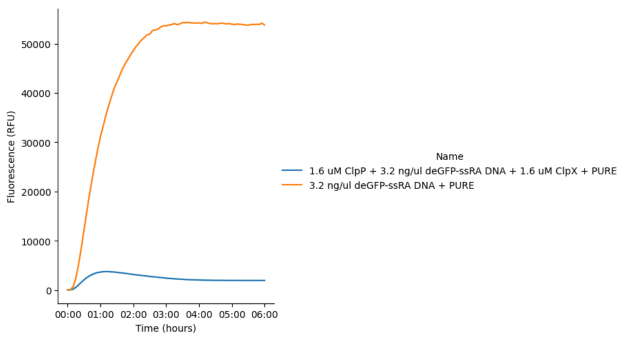
Fluorescence normalized by the control liposomes

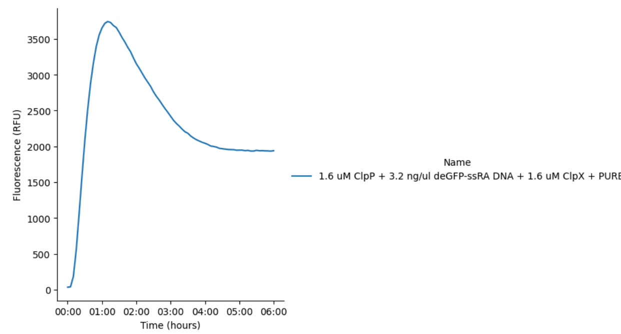
@October 15, 2025

	Sample 1 (C1	Sample 2	Sample 3
Purified ClpP (1.94 mg/ml)(79.9µM)	0.2	0	0
Linear ClpP (47.85 ng/ul)	0	0.5	0
Linear deGFP-ssRA (63.5 ng/ul)	0.5	0.5	0.5
Purified ClpX (2.55mg/ml)(53.7µM)	0.3	0.3	0
PURE	7.5	7.5	7.5
H2O	1.5	1.2	2
Total	10	10	10





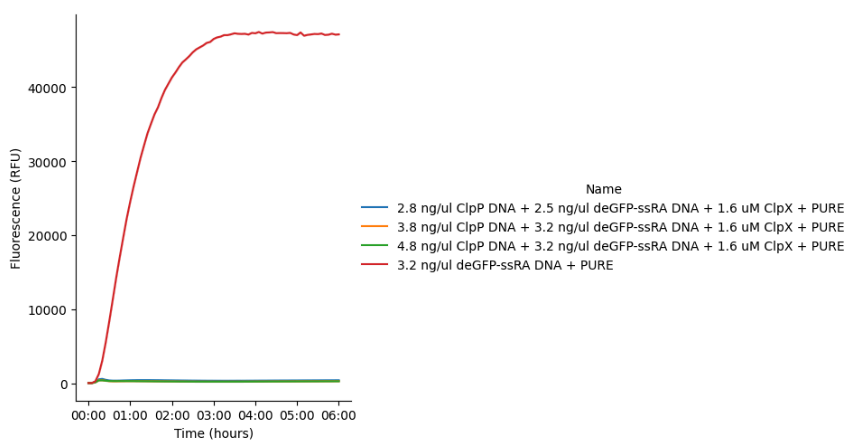


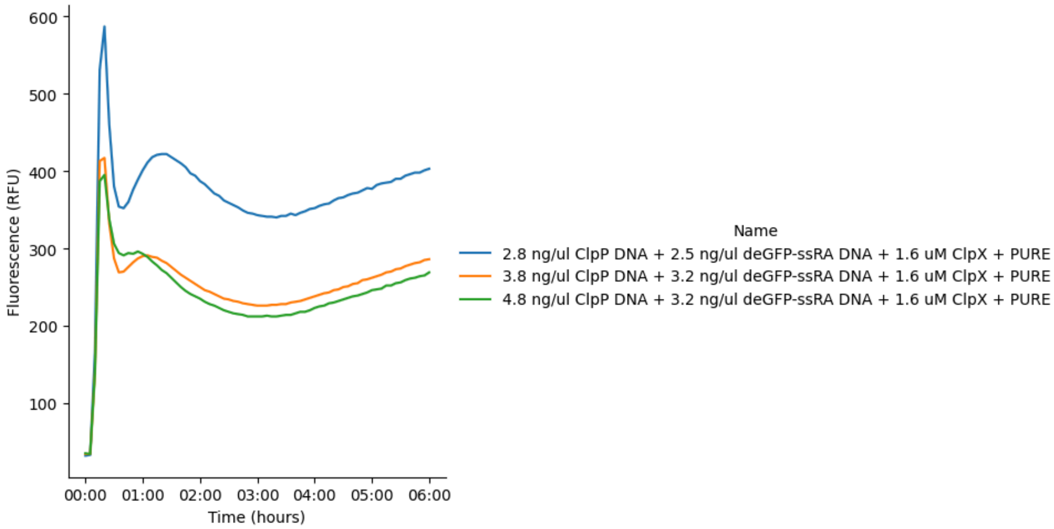


@October 16, 2025

Bulk Reactions:

	Sample 1 (D1	Sample 2	Sample 3	Control
Linear ClpP (47.85 ng/ul)	0.6	0.8	1	0
Linear deGFP-ssRA (63.5 ng/ul)	0.4	0.4	0.4	0.4
Purified ClpX (2.55mg/ml)(53.7µM)	0.4	0.4	0.4	0
PURE	7.5	7.5	7.5	7.5
H2O	1.1	0.9	0.7	2.1
Total	10	10	10	10

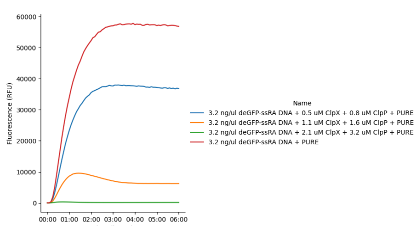


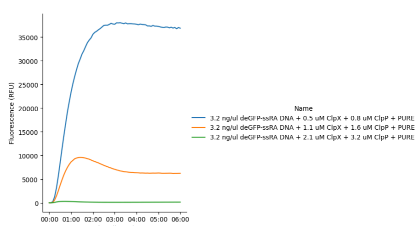


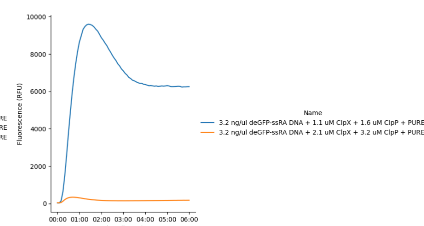
@October 20, 2025

Bulk Reactions:

	Sample 1 (F1	Sample 2	Sample 3	Control
Linear deGFP-ssRA (63.5 ng/ul)	0.5	0.5	0.5	0.5
Purified ClpP (1.94 mg/ml)(79.9µM)	0.4 (25% con.)	0.4 (50% con.)	0.4	0
Purified ClpX (2.55mg/ml)(53.7µM)	0.4 (25% con.)	0.4 (50% con.)	0.4	0
PURE	7.5	7.5	7.5	7.5
H2O	1.2	1.2	1.2	2
Total	10	10	10	10





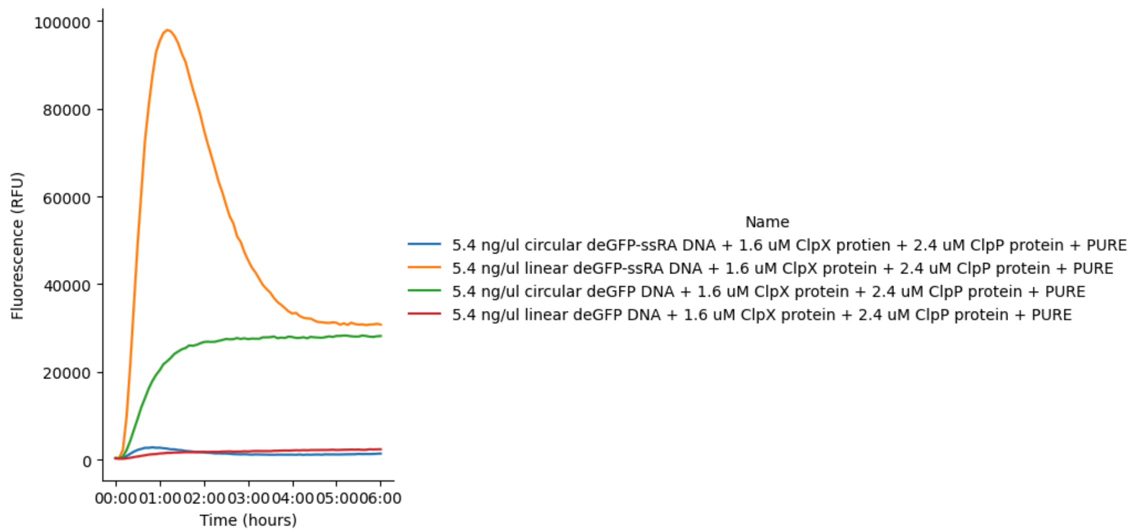


@November 6, 2025

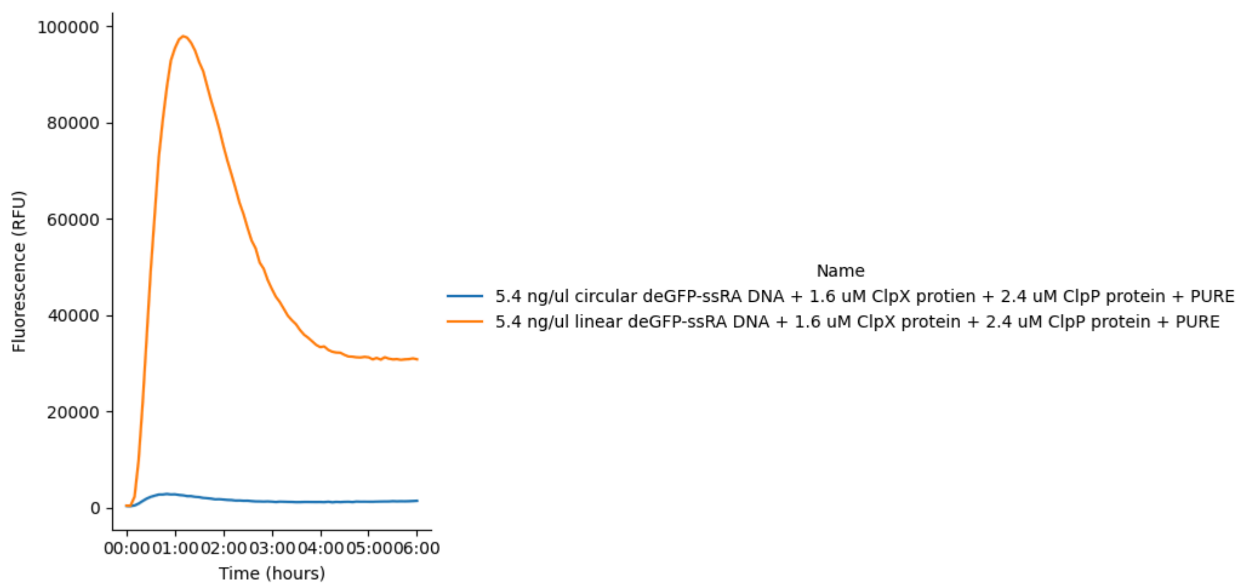
Bulk:

	Sample 1	Sample 2	Sample 3	Sample 4
pOpen-pT7-deGFP-ssrA (109.67 ng/ul)	0.5	x	x	x
Linear deGFP-ssrA (63.5 ng/ul)	x	0.86	x	x
pOpen-pT7-deGFP (74 ng/ul)	x	x	0.74	x
Linear pT7-deGFP (71 ng/ul)	x	x	x	0.77
Purified ClpP (1.94 mg/ml)(79.9µM)	0.3	0.3	0.3	0.3

Purified ClpX (2.55mg/ml)(53.7μM)	0.3	0.3	0.3	0.3
PURE	7.5	7.5	7.5	7.5
Optiprep	0.33	0.33	0.33	0.33
H2O	1.67	1.31	1.43	1.4
Total	10.6	10.6	10.6	10.6

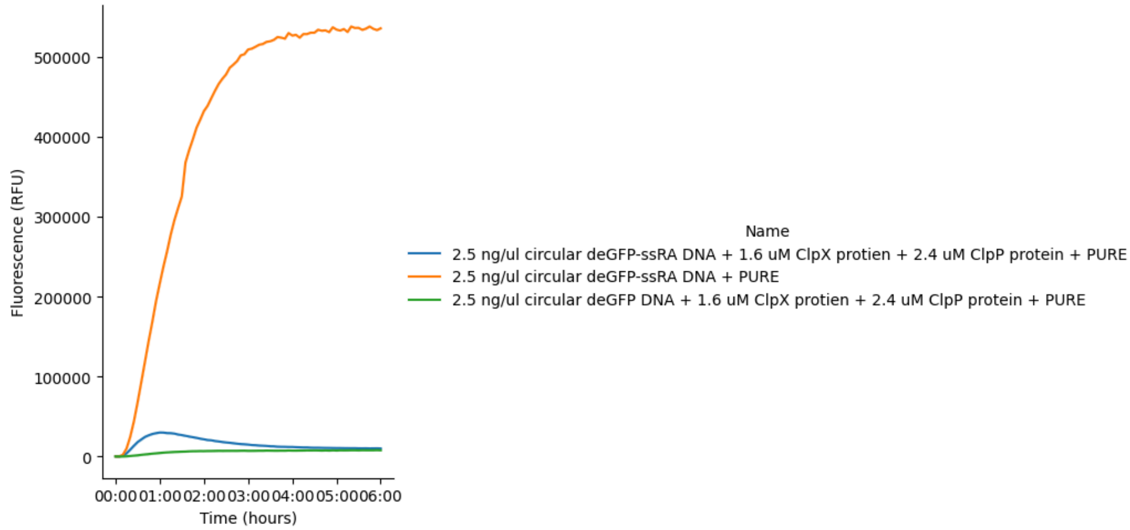


Linear deGFP-ssrA DNA works better than the circular one. Howwever, it seems like the qualities of the deGFP DNA stocks, either linrae one or the circular one, are not ideal.



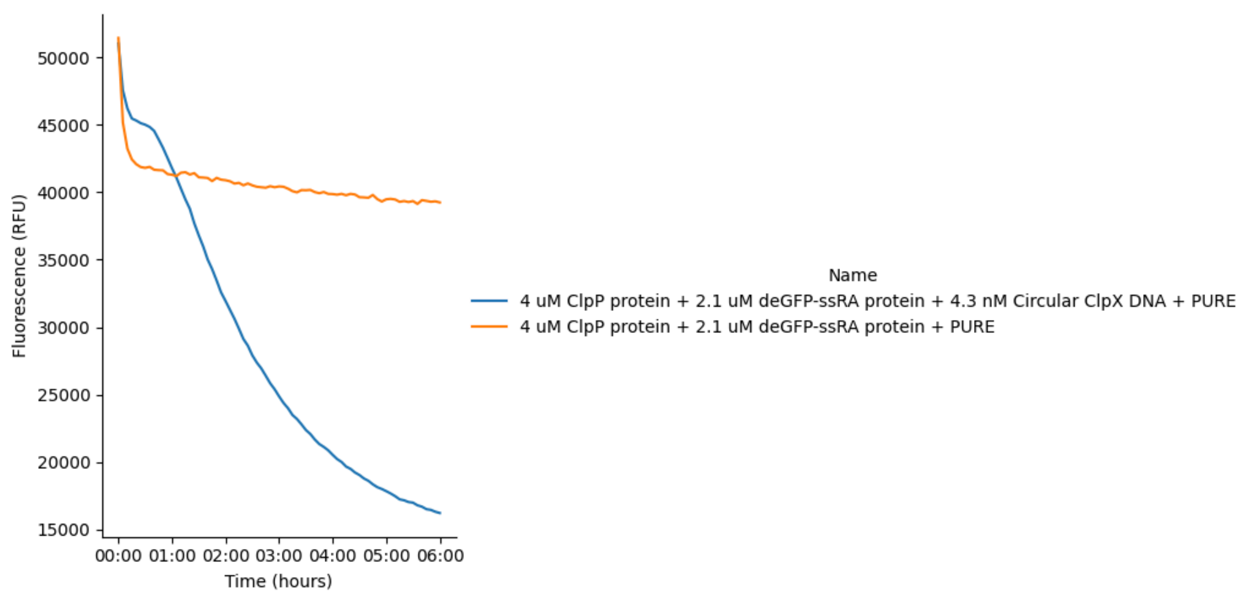
@November 14, 2025

Bulk reactions:



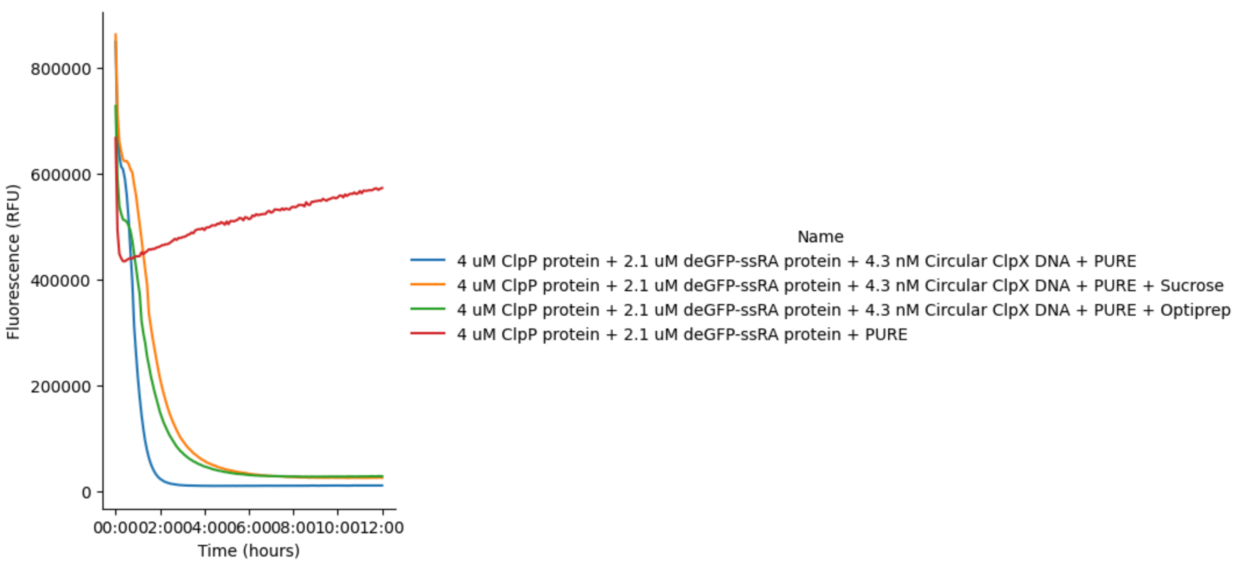
@December 5, 2025

	Sample	Control
Purified ClpP (1.94 mg/ml)(79.9 µM)	0.5	0.5
Purified deGFP-ssRA (1.15mg/ml)(41.2 µM)	0.5	0.5
pOpen-pT7-ClpX (86 nM)	0.5	0
PURE	7.5	7.5
H2O	1	1.5
Total	10	10



@December 9, 2025

	S1	S2	S3	Control
Purified ClpP (1.94 mg/ml)(79.9 µM)	0.5	0.5	0.5	0.5
Purified deGFP-ssRA (1.15mg/ml)(41.2 µM)	0.5	0.5	0.5	0.5
pOpen-pT7-ClpX (86 nM)	0.5	0.5	0.5	0
PURE	7.5	7.5	7.5	7.5
2M Sucrose	x	1	x	x
Optiprep	x	x	0.3	x
H2O	1	0	0.7	1.5
Total	10	10	10	10

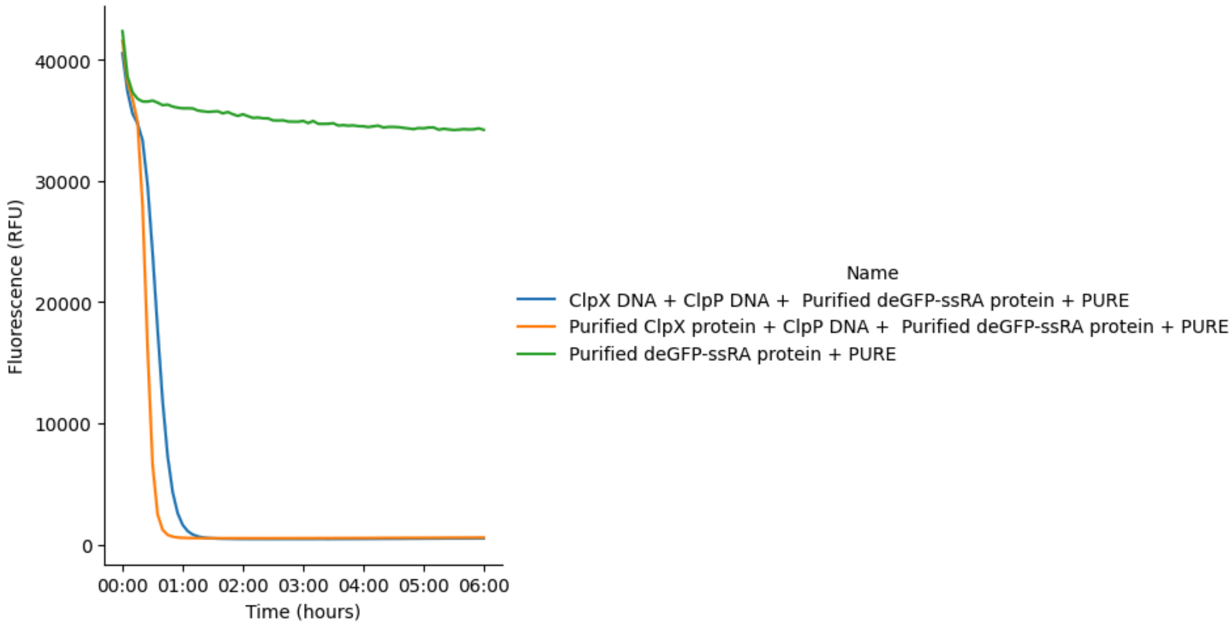


@December 11, 2025

Bulks-Test (Cy3):

	A1	A3	Control-A5
Linear pT7-ClpP (70 ng/ul)	0.5	0.5	x
Linear pT7-ClpX (70 hg/ul)	0.5	x	x
Circular pT7-ClpX (208 hg/ul)	x	x	x
Purified ClpP (79.9 µM)	x	x	x
Purified ClpX (53.7 ng/ul)	x	0.5	x
Purified deGFP-ssrA (41.2 ng/ul)	0.3	0.3	0.3
PURE	7.5	7.5	7.5
H2O	1.2	1.2	2.2

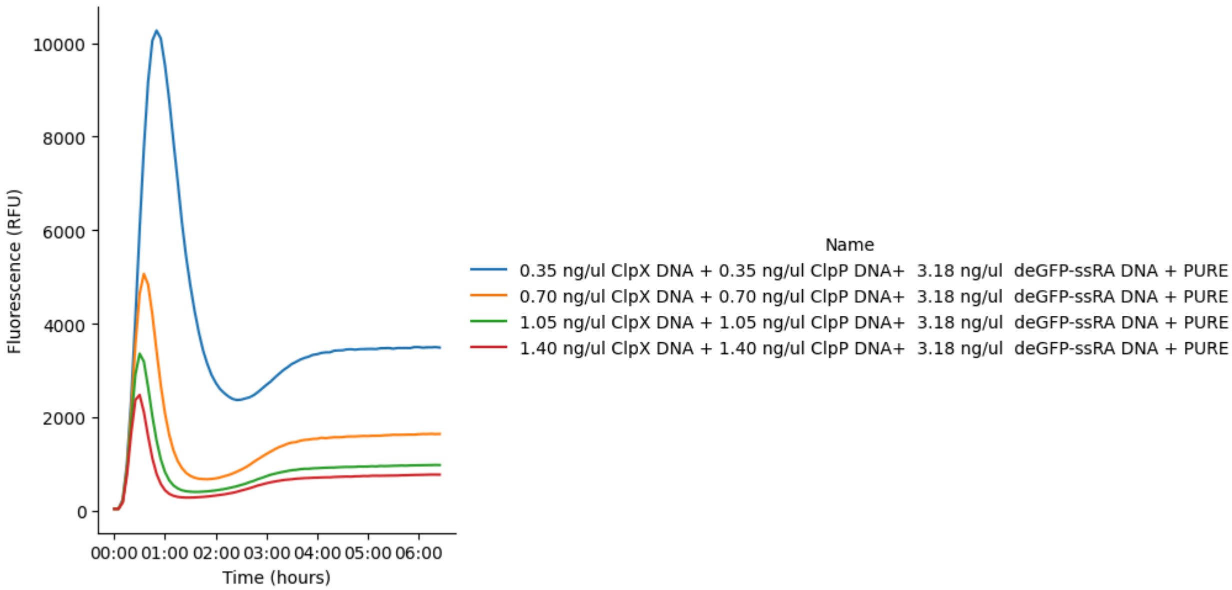
Total	10	10	10



@December 12, 2025

Bulks - Cy3:

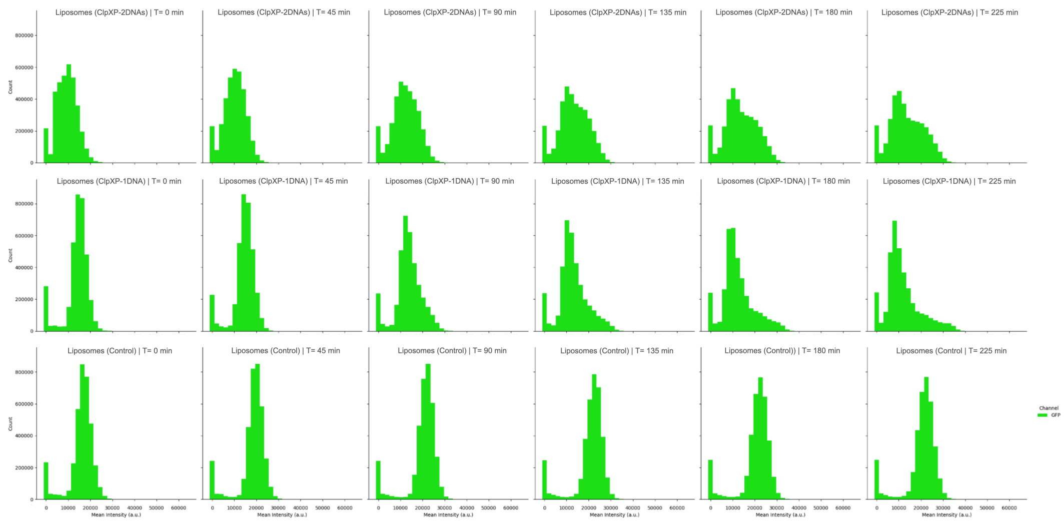
	B1	B2	B3	B4	Control-B5
Linear ClpP (17.5 ng/ul)	0.2	0.4	0.6	0.8	x
Linear deGFP-ssRA (63.5 ng/ul)	0.5	0.5	0.5	0.5	0.5
Linear pT7-ClpX (17.5 ng/ul)	0.2	0.4	0.6	0.8	x
PURE	7.5	7.5	7.5	7.5	7.5
H2O	1.6	1.2	0.8	0.4	2
Total	10	10	10	10	10



Cells:

	H11	H12	H13-Control
Linear pT7-ClpP (70 ng/ul)	1.5	1.5	x
Linear pT7-ClpX (70 hg/ul)	1.5	x	x
Purified ClpP (79.9 µM)	x	x	x
Purified ClpX (53.7 ng/ul)	x	0.5	x
Purified deGFP-ssrA (41.2 uM)	0.3	0.3	0.3
PURE	22.5	22.5	22.5
2M Glucose	3	3	3
H2O	1.2	2.2	4.2
Total	30	30	30

	H11	H12	H13-Control
Outer solution (Glucose)	933 mOsm	960 mOsm	900 mOsm


Liposomes (ClpXP-2DNAs) | T= 0 min
Liposomes (ClpXP-2DNAs) | T= 45 min
Liposomes (ClpXP-2DNAs) | T= 90 min
Liposomes (ClpXP-2DNAs) | T= 135 min
Liposomes (ClpXP-2DNAs) | T= 180 min
Liposomes (ClpXP-2DNAs) | T= 225 min
Liposomes (ClpXP-1DNA) | T= 0 min
Liposomes (ClpXP-1DNA) | T= 45 min
Liposomes (ClpXP-1DNA) | T= 90 min
Liposomes (ClpXP-1DNA) | T= 135 min
Liposomes (ClpXP-1DNA) | T= 180 min
Liposomes (ClpXP-1DNA) | T= 225 min
Liposomes (Control) | T= 0 min
Liposomes (Control) | T= 45 min
Liposomes (Control) | T= 90 min
Liposomes (Control) | T= 135 min
Liposomes (Control)) | T= 180 min
Liposomes (Control | T= 225 min
Count
Mean Intensity (a.u.)
Channel
GFP